

Supplemental Table S1: Modified Poisson regression models demonstrating the prevalence ratio (PR) of seeking urgent asthma care in the last 12 months for a given increase in pollutant concentration in **LAPN only**, stratified by presences vs. absence of dust mite allergen (derf1) in the settled home dust. The relationship between pollutants and urgent asthma visits was stronger among children with no dust mite (derf1) exposure.

	No dust mite (n=48)			Positive dust mite (n=43)		
	PR	95% CI	P	PR	95% CI	P
NO ₂	2.32	1.71, 3.15	<0.01	1.90	1.43, 2.54	<0.01
PM _{2.5}	14.4	2.93, 71.1	<0.01	10.6	3.24, 34.5	<0.01
EC	1.01	0.94, 1.08	0.81	1.08	1.04, 1.12	<0.01
O ₃	0.09	0.01, 0.79	0.03	0.57	0.32, 1.00	0.05
SO ₂	6.10	1.95, 19.0	<0.01	4.09	1.88, 8.88	<0.01

All models adjusted for sex, black race, maternal asthma, parental income, preventive asthma medication use, exercise-induced wheeze, building type, apartment unit floor, maternal hardship, and truck route density. Models not controlled for environmental tobacco smoke exposure, Latino ethnicity or white/Asian/mixed/other race because of lack of convergence of the models when these variables were added.

LAPN: lower asthma prevalence neighborhood; PR: prevalence ratio; CI: confidence interval; NO₂: nitrogen dioxide; PM_{2.5}: particulate matter <2.5 microns; EC: elemental carbon; O₃: ground level ozone; SO₂: sulfur dioxide

Supplemental Table S2: Modified Poisson regression models demonstrating the prevalence ratio (PR) of seeking urgent asthma care in the last 12 months for a given increase in pollutant concentration in **HAPN only**, stratified by the floor height of the apartment unit the child lived in.

	Floor 0-2 (n=36)			Floor 3-34 (n=63)		
	PR	95% CI	P	PR	95% CI	P
NO₂	1.42	0.92, 2.20	0.11	1.03	0.90, 1.17	0.70
PM_{2.5}	1.66	0.55, 5.01	0.37	1.12	0.61, 2.07	0.71
EC	1.00	0.97, 1.04	0.86	1.01	0.99, 1.04	0.15
O₃	1.00	0.73, 1.39	0.98	0.99	0.80, 1.23	0.93
SO₂	0.90	0.67, 1.22	0.52	0.92	0.71, 1.20	0.54

All models adjusted for sex, black race, Latino ethnicity, white/Asian/other/mixed race, preventive asthma medication use, exercise-induced wheeze, environmental tobacco smoke exposure, maternal asthma, parental income, building type, maternal hardship, and truck route density.

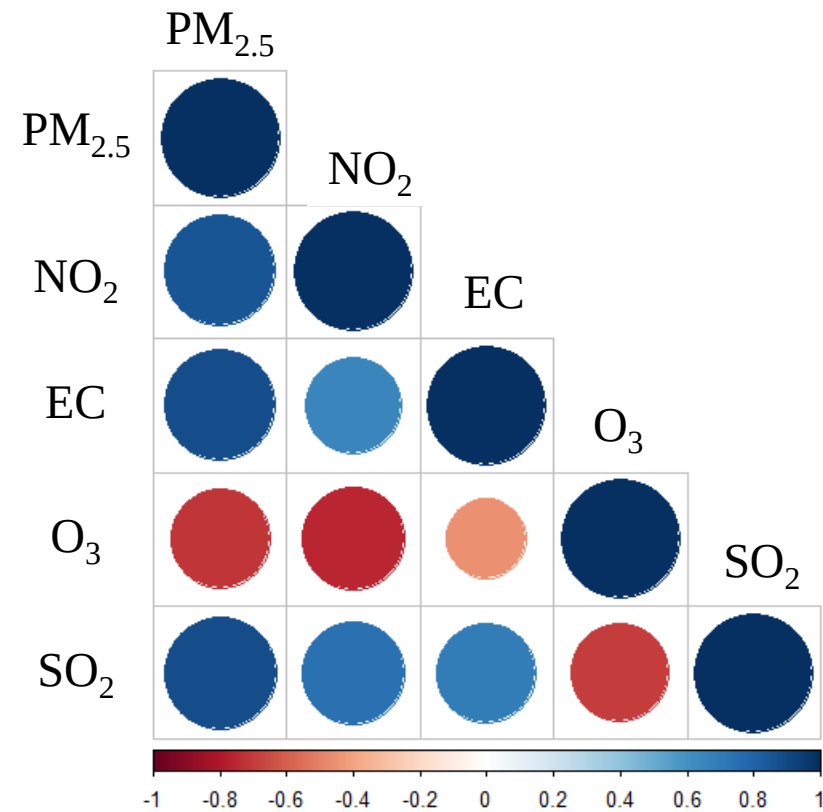
LAPN: lower asthma prevalence neighborhood; HAPN: higher asthma prevalence neighborhood; PR: prevalence ratio; CI: confidence interval; NO₂: nitrogen dioxide; PM_{2.5}: particulate matter <2.5 microns; EC: elemental carbon; O₃: ground level ozone; SO₂: sulfur dioxide

Supplemental Table S3: Modified Poisson regression models demonstrating the prevalence ratio (PR) of seeking urgent asthma care in the last 12 months for a given increase in pollutant concentration among the n=78 children that reported wheeze-related symptoms ≥ 4 times or sleep disturbance ≥ 1 time per week in the past 12 months.

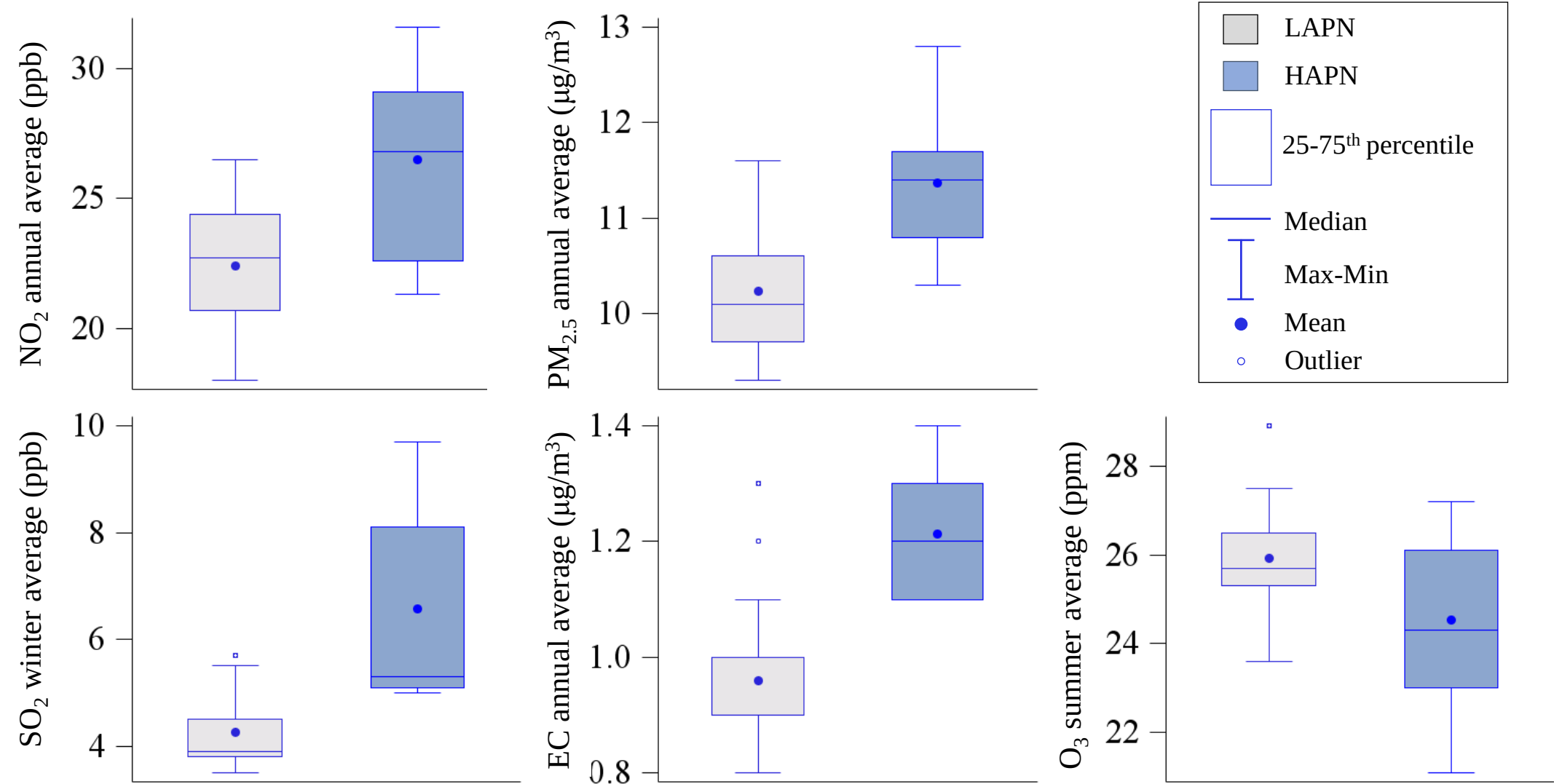
	Overall Sample (n=78)			LAPN (n=31)			HAPN (n=47)			
	PR	95% CI	P	PR	95% CI	P	PR	95% CI	P	P-interaction
NO ₂	1.04	0.97, 1.11	0.25	1.53	1.22, 1.92	<0.01	0.99	0.92, 1.06	0.73	0.02
PM _{2.5}	1.17	0.78, 1.74	0.45	2.68	0.88, 8.14	0.08	0.87	0.51, 1.50	0.63	0.07
EC	1.00	0.99, 1.02	0.57	1.00	0.95, 1.05	0.89	1.00	0.98, 1.03	0.95	-
O ₃	0.95	0.83, 1.08	0.43	0.48	0.35, 0.65	<0.01	1.08	0.92, 1.28	0.35	<0.01
SO ₂	1.00	0.84, 1.19	0.99	2.28	1.17, 1.47	0.02	0.92	0.73, 1.17	0.50	0.02

All models adjusted for sex, black race, Latino ethnicity, white/Asian/other/mixed race, preventive asthma medication use, exercise-induced wheeze, environmental tobacco smoke exposure, maternal asthma, parental income, building type, apartment unit floor height, maternal hardship, truck route density and neighborhood (only in non-stratified models). P-interaction represents the P value for the interaction term (pollutant x neighborhood).
LAPN: lower asthma prevalence neighborhood; HAPN: higher asthma prevalence neighborhood; PR: prevalence ratio; CI: confidence interval; NO₂: nitrogen dioxide; PM_{2.5}: particulate matter <2.5 microns; EC: elemental carbon; O₃: ground level ozone; SO₂: sulfur dioxide

Supplemental Figure 1: Spearman correlations across the 5 criteria pollutants. Blue color represents positive correlation and red color represents negative correlation. Darker colors indicate stronger associations.



Supplemental Figure 2: Box plots representing distribution of zip-code level pollutant concentration stratified by high asthma prevalence neighborhood (HAPN, n=91) and low asthma prevalence neighborhood (LAPN, n=99)



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DATE OF PREPARATION: June, 2018

PERSONAL DATA:

Name Stephanie Lovinsky-Desir
 Date of Birth July 25, 1981
 Birthplace Mount Vernon, NY
 Citizenship USA

WORK EXPERIENCE:

Academic Appointments

7/2013 – Present Division of Pediatric Pulmonology, Columbia University New York, NY
Assistant Professor of Pediatrics

Hospital Appointments

7/2013 – Present Division of Pediatric Pulmonology, NY Presbyterian Hospital New York, NY
Assistant Professor of Pediatrics

Other Work Experience

7/2013 – Present Division of Pediatric Pulmonology, Columbia University New York, NY
ICAC2 and ICAC3 Asthma Research Study Physician
Protocols APIC, PROSE, URECA, RACR, MUPPITS and CoNAC

EDUCATION:

8/2014 – 5/2016 Mailman School of Public Health, Columbia University New York, NY
 MS, Biostatistics, May 2016

8/2003 – 5/2007 New York Medical College Valhalla, NY
 MD, May 2007

8/1999 – 5/2003 LaSalle University Philadelphia, PA
 BA, Biology, May 2003

POST DOCTORAL TRAINING:

7/2010 – 7/2013 Division of Pediatric Pulmonology, Columbia University New York, NY
Pediatric Pulmonary Fellow

7/2007 – 6/2010 Department of Social Pediatrics, Albert Einstein - Montefiore Bronx, NY
Social Pediatrics Resident

7/2007 – 6/2008 Department of Social Pediatrics, Albert Einstein - Montefiore Bronx, NY
Social Pediatrics Intern

LICENSURE AND BOARD CERTIFICATION:

Licensure

2010 – Present New York State, License number: 258615 (active)

Board Certification

2014 – Present American Board of Pediatrics, Pulmonary Subspecialty, Certificate: 1158
 2010 – Present American Board of Pediatrics, Certificate: 98034

HONORS AND AWARDS:

- 2016 Society for Pediatric Research, elected member
- 2015 National Institute for General Medicine Scholarship Award – Epidemiology and Population Health Summer Institute at Columbia course on Global Information Systems
- 2012 Junior Investigator Award – Children’s Environmental Health Network Research Conference
- 2012 Assembly on Pediatrics Travel Award – American Thoracic Society
- 2011 Comparative Effectiveness Research – Summer Training in Applications and Techniques Scholarship Award for the Columbia Summer Research Institute (CSRI)
- 2010 Chairman’s Award for Distinction in Research – Social Medicine at Montefiore
- 2010 Minority Trainee Travel Award Recipient – American Thoracic Society
- 2009 Resident Travel Award Recipient – American Thoracic Society
- 2007 Dr. Lester J. Schultz Memorial Scholarship Award – New York Medical College

PROFESSIONAL ORGANIZATIONS, SOCIETIES AND SERVICE:Memberships and Positions

- 2016 – Present American Thoracic Society, Assembly on Pediatrics Program Committee
- 2017 – Present American College of Chest Physicians, Diversity Task Force
- 2015 – 2016 American College of Chest Physicians, Diversity Committee
- 2015 – Present Pediatric Pulmonology Program Assessment Committee at Columbia University Medical Center
- 2014 – Present American College of Chest Physicians, *member*
- 2012 – 2014 Academic Pediatric Association, *member*
- 2007 – 2010 American Academy of Pediatrics, *member*
- 2007 – Present American Thoracic Society, *member*

Journal Reviewer

- 2017 – Present Environment International, *ad hoc reviewer*
- 2016 – Present Clinical Epigenetics, *ad hoc reviewer*
- 2015 – Present Chemosphere, *ad hoc reviewer*
- 2015 Yearbook of Pediatrics, *contributing editor*

FELLOWSHIP AND GRANT SUPPORT:Present Support

- 7/2018 – 6/2019 John M. Driscoll, Jr. MD. Children’s Scholar Fund Lovinsky-Desir (PI)
Title: Adolescent Physical Activity in Urban Polluted Environments and Respiratory Measures
Role: Principal Investigator
- 4/2018 – 3/2020 5P30 ES009089 A. Bacarelli (PI)
 NIEHS Center Career Development Award
Title: Air Pollution and Asthma in NYC Schools: How can we reduce exposure during activity?
Role: Principal Investigator (individual project)
- 4/2018 – 3/2022 K01HL140216 S. Lovinsky-Desir (PI)
Title: Physical Activity and Dose of Air Pollution in Pediatric Urban Asthma: Impact of Minute Ventilation
Role: Principal Investigator

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7/2017 – 6/2021	Amos Medical Faculty Development Award Robert Wood Johnson Foundation <i>Title:</i> Exercise in urban polluted environments and childhood asthma <i>Role:</i> Principal Investigator	S. Lovinsky-Desir (PI)
7/2017 – 6/2019	L40 ES023169-03 NIH Extramural Loan Repayment Program for Pediatric Research <i>Title:</i> Childhood Asthma and the Impact of Exercise in Urban Polluted Environments <i>Role:</i> Principal Investigator	S. Lovinsky-Desir (PI)
<u>Past Support</u>		
9/2015 – 8/2017	KL2 TR000081 TRANSFORM KL2 Mentored Career Development Award Irving Institute for Clinical and Translational Research <i>Title:</i> Influence of Physical Activity Level and Air Pollution Exposure in Asthma <i>Role:</i> Principal Investigator (individual project)	Melissa Begg (PI)
7/2015 – 12/2016	John M. Driscoll, Jr. MD. Children's Scholar Fund <i>Title:</i> Adolescent Physical Activity in Urban Polluted Environments and Respiratory Measures <i>Role:</i> Principal Investigator	Lovinsky-Desir (PI)
2/2015 – 6/2016	Columbia University Provost's Grants Program for Junior Faculty <i>Title:</i> The impact of physical activity level on environmental and epigenetic mechanisms in asthma <i>Role:</i> Principal Investigator	
9/2014 – 12/2015	CHEST – Diversity Young Investigator Award <i>Title:</i> Urban tree canopy exposure, DNA methylation, and allergies in pediatric asthma <i>Role:</i> Principal Investigator	Lovinsky-Desir (PI)
9/2013 – 10/2015	3R01ES013163-07S1 NIH Supplement to Promote Diversity in Health Related Research <i>Title:</i> Black carbon exposure, DNA methylation, airway inflammation in pediatric asthma <i>Role:</i> Research fellow	Rachel L. Miller (PI)
7/2013 – 6/2014	Stony Wold – Herbert Fund Fellowship <i>Title:</i> Black carbon exposure and DNA methylation in pediatric asthma: modification by obesity, activity <i>Role:</i> Principal Investigator	Lovinsky-Desir (PI)
7/2015 – 6/2016	L40 ES023169-02 NIH Extramural Loan Repayment Program for Pediatric Research <i>Title:</i> Physical Activity Level, Environment and Epigenetics in Asthma <i>Role:</i> Principal Investigator	Lovinsky-Desir (PI)

- 7/2011 – 6/2013 5T32HL105323-02 Jahar Bhattacharya (PI)
 NIH T32 Institutional National Research Service Award – Training Grant
Role: Research fellow/Trainee
- 7/2013 – 6/2015 L40 ES023169-01 Lovinsky-Desir (PI)
 NIH Extramural Loan Repayment Program for Pediatric Research
Title: BC exposure & epigenetics in pediatric asthma: Modification by obesity activity
Role: Principal Investigator

PUBLICATIONS:

Peer Reviewed Publications

- Lovinsky-Desir S**, Lawrence J, Jung KH, Rundle AG, Hoepner LA, Yan B, Perera F, Perzanowski MS, Miller RL, Chillrud SN. Air Pollution, Assessment of exposure to air pollution in children: Determining whether wearing a personal monitor affects physical activity. *Environmental Research*. 2018, 166:340-3.
- Lovinsky-Desir S**, Jung KH, Jezioro JR, Torrone DZ, dePlanell-Saguer M, Yan B, Perera FP, Rundle AG, Perzanowski MS, Chillrud SN, Miller RL. Physical Activity, Black Carbon Exposure and DNA Methylation in the FOXP3 Promoter. *Clinical Epigenetics*. 2017, 9: 65.
- Jung KH, **Lovinsky-Desir S**, Yan B, Torrone D, Lawrence J, Jezioro JR, Perzanowski M, Perera FP, Chillrud SN, Miller RL. Effect of personal exposure to black carbon on changes in allergic asthma gene methylation measured 5 days later in urban children: importance of allergic sensitization. *Clinical Epigenetics*. 2017, 9: 61.
- Miller RL, Zhang H, Jezioro J, De Planell Saguer M, **Lovinsky-Desir S**, Liu X, Perzanowski M, Divjan A, Phipatanakul W, Matsui E. Reduced mouse allergen is associated with epigenetic changes regulatory genes, but not mouse sensitization, in asthmatic children. *Environmental Research*. 2017, 156: 619-24.
- Jung KH, Torrone D, **Lovinsky-Desir S**, Perzanowski M, Bautista J, Jezioro JR, Hoepner L, Ross J, Perera P, Chillrud SN, Miller RL. Short-term exposure to PM_{2.5} and vanadium and changes in asthma gene DNA methylation and lung function decrements among urban children. *Respiratory Research*. 2017, 18(1): 63.
- Lovinsky-Desir S**, Jung KH, Rundle AG, Hoepner L, Bautista J, Perera FP, Chillrud SN, Perzanowski MS, Miller RL. Physical Activity is Associated with Greater Black Carbon/Soot Exposure and Lower Fractional Exhaled Nitric Oxide in a Cohort of Adolescents Living in New York City. *Environmental Research*. 2016, 151: 756-62.
- Lovinsky-Desir S**, Miller RL, Bautista J, Gil EN, Chillrud SH, Yan B, Perera FP, Jung KH. Differences in ambient polycyclic aromatic hydrocarbon concentrations between streets and alleys in New York City: Open space vs semi-closed space. *Int. J. Environ. Res. Public Health*. 2016, 13: 127.
- Jung KH, **Lovinsky-Desir S**, Perzanowski M, Liu X, Maher C, Gil E, Torrone D, Sjodin A, Li Z, Perera FP, Miller RL. Repeatedly high polycyclic aromatic hydrocarbon exposure and cockroach sensitization among inner-city children. *Environment Research*; 2015, 140: 649-56.
- Lovinsky-Desir S**, Ridder R, Torrone D, Maher C, Narula S, Scheuerman M, Kattan M, DiMango E, Miller RL. DNA methylation of the allergy regulatory gene, interferon gamma, varies by age, sex, and tissue type in asthmatics. *Clinical Epigenetics*; 2014, 6(9).

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- Jung KH, Lui B, **Lovinsky-Desir S**, Yan B, Camann D, Sjodin A, Li Z, Perera F, Kinney P, Chillrud S, Miller RL. Time trends of polycyclic aromatic hydrocarbon exposure in New York City from 2001 to 2012: Assessed by repeat air and urine samples. *Environmental Research*; 2014, 131: 95-103.
- Lovinsky-Desir S**, Folch C, Jung KH, Torrone D, Gil E, Perera F, Miller RL, Chillrud SN. Urban adolescents readily comply with a complicated asthma research protocol. *Clinical Medicine Insights: Circulatory, Respiratory and Pulmonary Medicine*; 2014, Mar 23 (8):5-9.
- Lovinsky S**, Rastogi D. Prescription habits for preventative medications among pediatric emergency department physicians at an inner city teaching hospital. *Journal of Asthma*; 2010, 47(9): 1011-14.

Reviews

- Phinizy PA, **Lovinsky-Desir S**. 3 Practical Tests of Pulmonary Function in Preschool Children. *Chest Magazine: Pulmonary Perspectives*. 2016, April 11(4): 12.
- Lovinsky-Desir S**, Bye MR. Laryngomalacia. *Medscape Reference*. Updated March 28, 2014. Available at: <http://emedicine.medscape.com/article/1002527-overview>.
- DePlanell Sagner M, **Lovinsky-Desir S**, Miller RL. Epigenetic regulation: The interface between prenatal and early-life exposure and asthma susceptibility. *Environmental and Molecular Mutagenesis*; 2014, Apr; 55(3): 231-43.
- Lovinsky-Desir S**, Miller RL. Epigenetics, asthma and allergic diseases: A review of latest advancements. *Current Allergy and Asthma Reports*; 2012, Jun; 12(3):211-20.

Book Chapters

- Lovinsky-Desir, S.** and Miller, R. L. (2012) Environmental Epigenetics, Asthma, and Allergy: Our Environment's Molecular Footprints, in Toxicology and Epigenetics (ed S. C. Sahu), John Wiley & Sons, Ltd, Chichester, UK. doi: 10.1002/9781118349045.ch9

Case Reports

- Panghaal V, Ortiz-Romero S, **Lovinsky S**, Levin T. Benign acute childhood myositis; an unusual cause of calf pain. *Pediatric Radiology*; 2008, 38(6): 703-5.

Abstracts

- Lovinsky-Desir S**, Miller RL, Chillrud SN, Perzanowski MS, Jung KH. Timing and Mode of Commute Impacts in-Transit Black Carbon Exposure and Airway Inflammation. *American Thoracic Society International Meeting*. May 2018 (Oral presentation).
- Lovinsky-Desir S**, Acosta LM, Rundle AG, Miller RL, Goldstein IF, Jacobson JS, Chillrud SN, Perzanowski MS. Air Pollution, Urgent Asthma Care, and the Impact of Neighborhood Asthma Prevalence. *CHEST*. October 2017; 152 (4): A829 (Oral presentation).
- Lovinsky-Desir S**, Jung KH, Rundle AG, Beizhan Y, Chillrud SN, Perera FP, Perzanowski MS, Miller RL. The Relationship Between Physical Activity and Lung Function Is Influenced by Black Carbon Pollutant Exposure. *American Thoracic Society International Meeting. Am J Respir Crit Care Med*. 2017; 195: A3899.
- Lovinsky-Desir S**, Lovasi G, Liu X, Torrone D, Jung KH, Jezioro J, Perera FP, Rundle AG, Miller RL. Greater Urban Tree Canopy Exposure May Be Linked to Asthma and DNA Methylation of the IL4 Gene Promoter. *American Thoracic Society International Meeting. Am J Respir Crit Care Med*. 2016. 193; 2016: A2929.

- Lovinsky-Desir S**, Torrone D, Jung KH, Gil E, Bautista J, Perera F, Perzanowski M, Miller RL. DNA demethylation of the inducible nitric oxide synthase gene is associated with decreased lung function in children. *Am J Respir Crit Care Med*.191; 2015: A3493.
- Lovinsky-Desir S**, Ridder R, Torrone D, Zhang H, Merle D, Narula S, Kattan M, DiMango E, Miller RL. Interferon gamma promoter DNA methylation is correlated with age and sex in a cohort of allergic sensitized adults and children with asthma. *Am J Respir Crit Care Med*.189; 2014: A3385.
- Jung KH, Lui B, **Lovinsky-Desir S**, Yan B, Camann D, Perera F, Kinney P, Chillrud S, Miller RL. Diverging time trends of polycyclic aromatic hydrocarbon, black carbon and fine particulate matter exposures in New York City. *Am J Respir Crit Care Med*.189; 2014: A3151.
- Lovinsky-Desir S**, Ridder R, Torrone D, Jaing H, Narula S, Kattan M, DiMango E, Miller RL. IFN γ promoter DNA methylation levels in peripheral CD4+ lymphocytes versus buccal cells. *Am J Respir Crit Care Med*. 187; 2013: A1204.
- Lovinsky-Desir S**, Ridder R, Torrone D, Jaing H, Narula S, Kattan M, DiMango E, Miller RL. Epigenetic regulation of interferon (IFN)-gamma following a multifaceted environmental intervention in a pediatric and adult cohort of asthmatics. Children's Environmental Health Network 2012 Research Conference. The Contribution of Epigenetics in Pediatric Environmental Health, San Francisco, CA, 2012 (*Junior Investigator Award recipient*).
- Lovinsky-Desir S**, Donohue KM, Perzanowski MS, Just AC, Canfield S, Resnick D, Sheares BJ, Perera F, Whyatt R, Kattan M, Miller RL. Standardization of the diagnosis of current asthma in pediatric cohort studies and its association with the presence or absence of bronchodilator reversibility. *Am J Respir Crit Care Med*. 185; 2012: A3367.
- Lovinsky-Desir S**, Ovchinsky N, Kattan M, Mellins RB, Sheares BJ. Abernathy malformation: an unusual diagnosis presenting as progressive hypoxemia in a child with underlying pulmonary pathology. *Am J Respir Crit Care Med*. 185; 2012: A6913.
- Jung, Kyung H, **Lovinsky-Desir S**, Liu B, Folch C, Perzanowski M, Yan B, Chillrud S, Whyatt R, Camann D, Kinney P, Perera F, and Miller RL. Characterization of polycyclic aromatic hydrocarbons levels in repeat indoor air and urine samples in a prospective birth cohort study. International Society of Exposure Science conference. 2012 (late breaking abstract).
- Cornell AG, **Lovinsky-Desir S**, Borczuk A, Mellins RB, Sheares BJ. CPAP in the management of a 5-month-old with late clinical presentation of primary pulmonary lymphangectasia. *Am J Respir Crit Care Med*. 183; 2011: A1880.
- Lovinsky S**, Langston-Davis N. Parental perceptions of the "Asthma Health Passport," a tool designed to track asthma treatment in acute and maintenance care settings. *Am J Respir Crit Care Med*. 181; 2010: A2736.
- Lovinsky S**, Rastogi D. An electronic medical record review quantifying prescription of controller medications to children with asthma in the pediatric emergency department. *Am J Respir Crit Care Med*. 181; 2010: A6646.

EDUCATIONAL CONTRIBUTIONS:

Direct Teaching/Precepting/Supervising

- 2013 – present Participation in Pediatric Pulmonology fellowship training of 3 fellows per year including teaching in bi-weekly conferences, precepting outpatient clinic one session every other month and inpatient service attending 6 weeks per year.

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Advising and Mentorship

- 2014 – 2017 Faculty Advisor and scholarly Oversight Committee member for Pediatric Pulmonology fellow, Khalil Savary, MD
Current Status: Assistant Professor of Pediatrics, Rutgers-NJ Medical School
- 2016 – present Scholarly Oversight Committee Member for Pediatric Pulmonology fellow, Theodosia Thoma, MD

INVITED LECTURES, PRESENTATIONS, AND RESEARCH SEMINARS:International

July 2017: Association of Haitian Physicians Abroad, Côtes des Arcadins, Haiti – “Pediatric Asthma in Inner Cities”

Regional

November 2017: One Health Symposium, Syracuse, NY – “Childhood Physical Activity and the Risk of Air Pollution Exposure in pediatric Asthma”

May 2017: Group Against Smog and Pollution - Making the Connection, Pittsburgh, PA – “Physical Activity, Air Pollution, and Asthma in the Urban Environment”

May 2017: University of Pittsburgh Environmental and Occupational Health Departmental Seminar, Pittsburgh, PA – “Activity and asthma in the urban environment: Does air pollution offset the respiratory benefit of exercise?”

Local

January 2018: Columbia University Department of Pediatrics Grand Rounds, New York, NY – “Physical Activity in Polluted Environments and the Impact on Pediatric Lungs”

October 2017: Maimonides Medical Center, Department of Pediatrics Grand Rounds, Brooklyn, NY – “The Benefits of Physical Activity in Childhood Asthma: Does Air Pollution Play a Role?”

June 2016: Columbia University Department of Pediatrics Chief of Service, New York, NY – “Exercise Induced/Reduced Asthma: Keeping Children with Asthma Active”